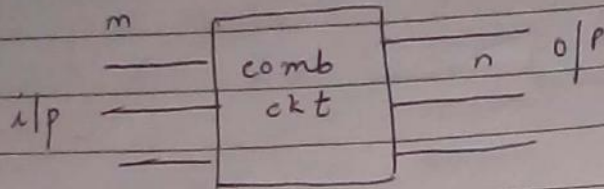


Unit-2: COMBINATIONAL CIRCUIT

In combinational circuit, output depends upon only present state not on past output and no feedback path is connected from output to input.



DESIGN procedure to analyse any COMBINATIONAL CIRCUIT

Step 1: Determine no. of i/p and no. of o/p.

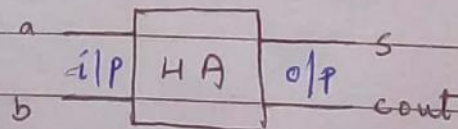
Step 2: Draw the truth table

Step 3: Now with the help of Boolean Algebra or K-Map, determine minimized value of every output which is f^n of i/p

Step 4: Now with the help of Logic Gates draw every o/p w.r.t i/p

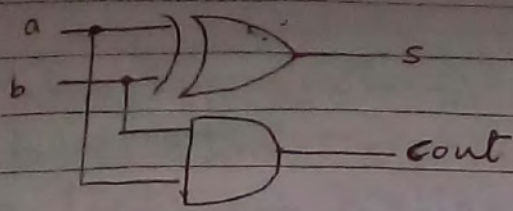
HALF ADDER

i/p		o/p	
a	b	s	cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



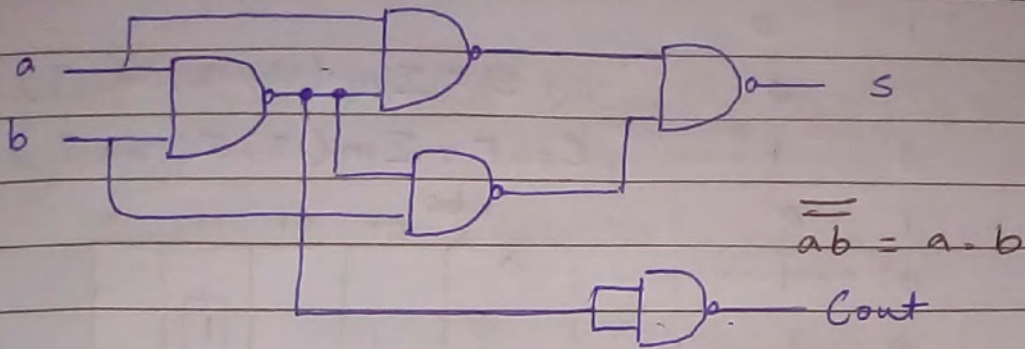
$$s = a \oplus b$$

$$\text{cout} = a \cdot b$$



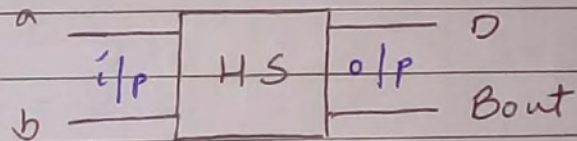
Q. Implement half adder by using min. no. of NAND gates.

5 Gates



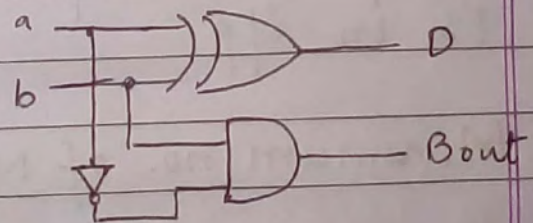
HALF SUBTRACTOR

i/p		o/p	
a	b	D	Bout
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



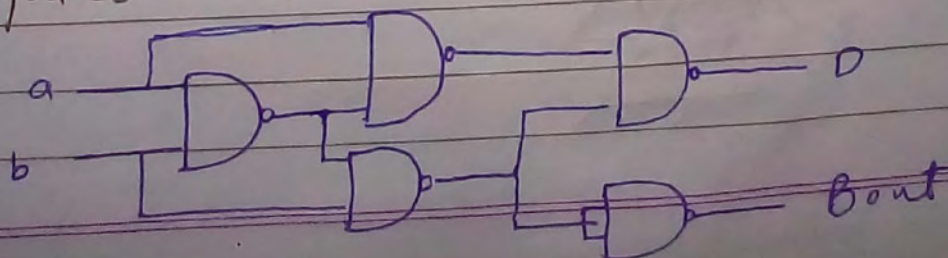
$$D = a \oplus b$$

$$Bout = \bar{a}.b$$



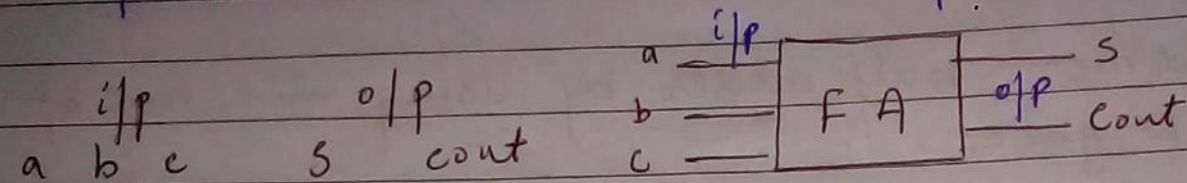
Q. Implement half subtractor by using min. no. of NAND gates

5 Gates



FULL ADDER

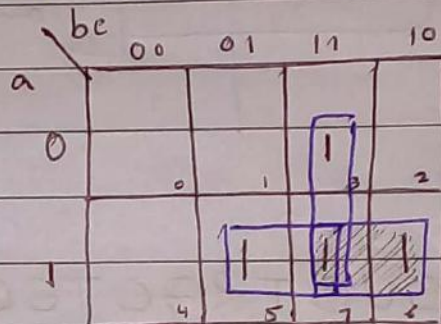
In half adder, it does not add carry, it simply adds 2 single bits and this problem is eliminated by full adder.



i/p			o/p	
a	b	c	S	Count
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$S = \sum m(1, 2, 4, 7) = a \oplus b \oplus c$$

$$\text{Count} = \sum m(3, 5, 6, 7) = ab + bc + ca$$

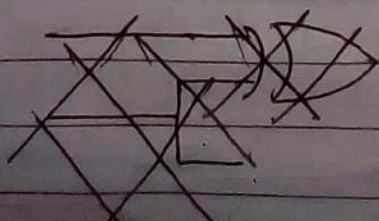


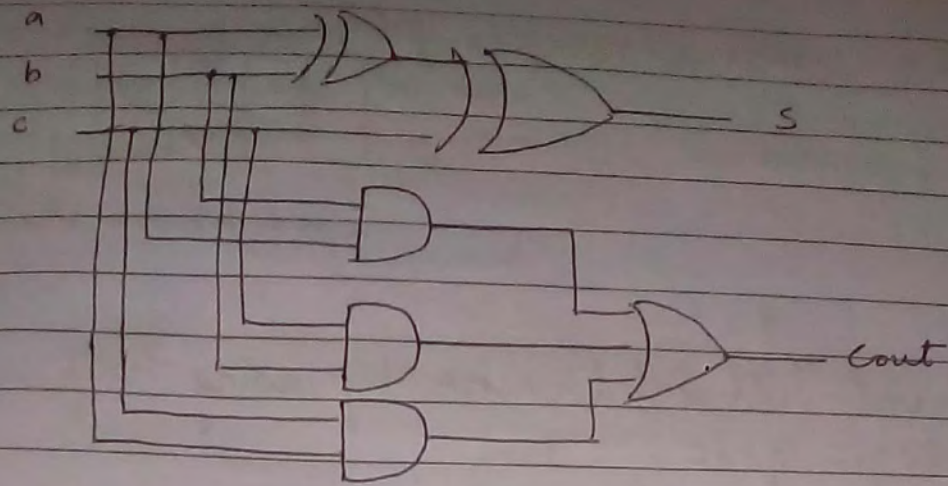
$$bc + ac + ab$$

* sum of full adder will be high, when total no. of ~~one~~ 1's at i/p are odd. Hence XOR expⁿ.

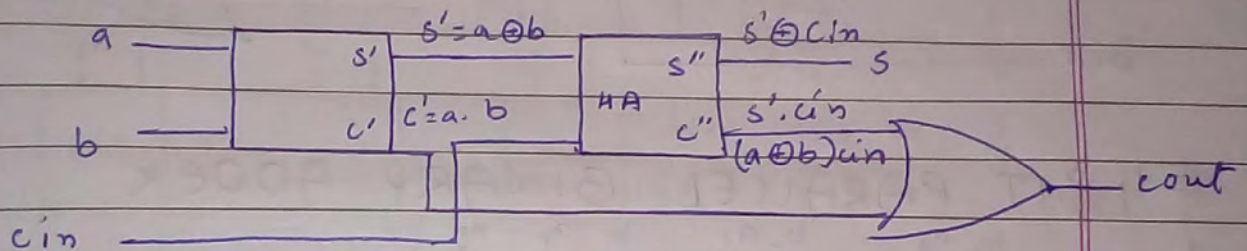
* Count will be high, when there are two or more 1's in i/p.

Minimum no. of NAND gates = 9





Q. implement full adder by using half adder and logic gates.



$$s = a \oplus b \oplus c_{in}$$

$$cout = \sum m(3, 5, 6, 7)$$

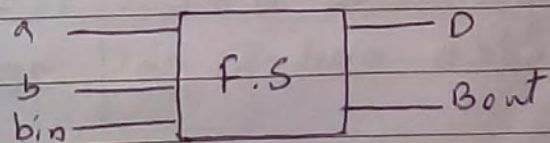
$$011 \ 101 \ 110 \ 111$$

$$\begin{aligned} cout &= \bar{a}bc_{in} + a\bar{b}c_{in} + ab\bar{c}_{in} + abc_{in} \\ &= (\bar{a}b + \bar{b}a)c_{in} + ab(\bar{c}_{in} + c_{in}) \\ &= (a \oplus b)c_{in} + ab \end{aligned}$$

FULL SUBTRACTOR

$a - (b + bin)$

a	b	bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



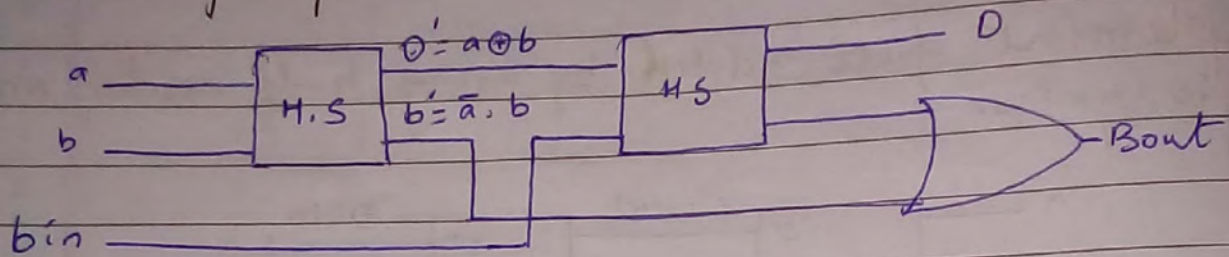
$$\begin{aligned} D &= \sum m(1, 2, 4, 7) \\ &= a \oplus b \oplus bin \end{aligned}$$

$$\begin{aligned} Bout &= \sum m(1, 2, 3, 7) \\ &= \bar{a}b + bbin + bin\bar{a} \end{aligned}$$

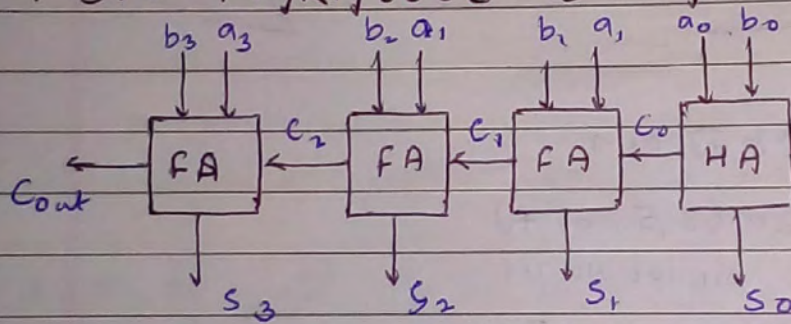
	00	01	11	10
a		1	1	1
b		1	1	1

$$\bar{a}b + b\bar{b}in + b\bar{a}in$$

Q. Implement full subtractor by using half subtractor and logic gate.



4 BIT PARALLEL BINARY ADDER



$$\begin{array}{r}
 \text{Cout} \quad \text{C}_2 \quad \text{C}_1 \quad \text{C}_0 \\
 a_3 \ a_2 \ a_1 \ a_0 \\
 b_3 \ b_2 \ b_1 \ b_0 \\
 \hline
 s_3 \ s_2 \ s_1 \ s_0
 \end{array}$$

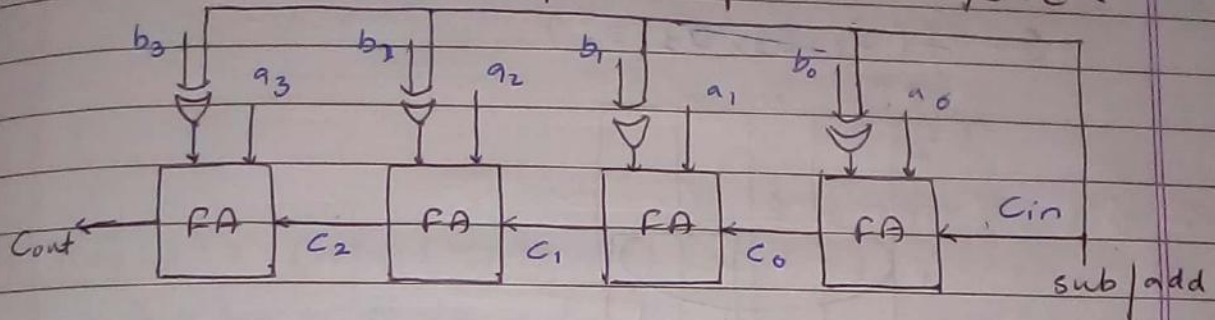
Imp.

* To implement n-bit parallel adder we require (n-1) full adder and 1 half adder or n full adders

4 BIT PARALLEL BINARY SUBTRACTOR



4 BIT PARALLEL ADDER / SUBTRACTOR



If $sub/add = 0$, it works as 4 bit 11el adder
 $= 1$, it works as 4 bit 11el sub

Q. Minimum no. of half adder and or gates to implement 8 bit 11el subtracter.

8 FA $\Rightarrow$ 16 HA + 8 OR Gate

BCD ADDER

In this if $sum > 9$ add 6

S/(cout)	S_3	S_2	S_1	S_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	0	1	0
5	0	0	1	1
6	0	0	1	0
7	0	0	1	1

8	0	1	0	0	0
9	0	1	0	0	1
10	1	0	0	0	0
11	1	0	0	0	1
12	1	0	0	1	0
13	1	0	0	1	1
14	1	0	1	0	0
15	1	0	1	0	1

$$S = \sum m(10, 11, 12, 13, 14, 15)$$

$$S = f(S_3, S_2, S_1, S_0)$$

	$S_1 S_0$	00	01	11	10
$S_3 S_2$	00				
	01				
	11	1	1	1	1
	10			1	1

$$S = S_3 S_2 + S_3 S_1$$

1st 11cl
adder

$a_3 \ a_2 \ a_1 \ a_0$

$b_3 \ b_2 \ b_1 \ b_0$

$c_0 \ S_3 \ S_2 \ S_1 \ S_0$

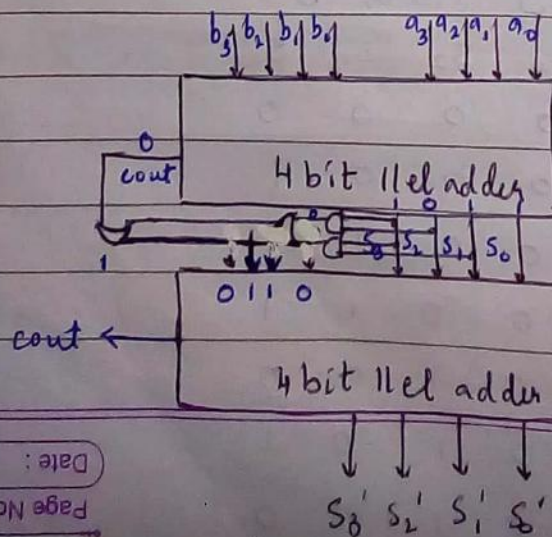
2nd 11cl
adder

0 0 0 0

0 1 1 0

cout $S'_3 \ S'_2 \ S'_1 \ S'_0$

if $a_4 \leq 9$ or $c_0 = 0$
if > 9 or $c_0 = 1$



3 BIT PARITY GENERATOR

We check at output either there is even parity or odd parity

a	b	c	$P_G(\text{even})$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$P_G = \sum m(1, 2, 4, 7)$$

$$= a \oplus b \oplus c$$



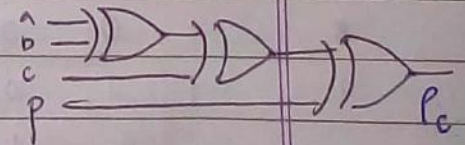
PARITY CHECKER

1 Parity bit is added with message signal and we check whether there is any error or not at output for even parity or odd parity.

a	b	c	p	$P_C(\text{even})$
0	0	0	0	0
0	0	0	1	1 ✓
0	0	1	0	1 ✓
0	0	1	1	0
0	1	0	0	0 ✓
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1 ✓
1	0	0	0	1 ✓
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1 ✓
1	1	0	0	0
1	1	0	1	1 ✓
1	1	1	0	0
1	1	1	1	1 ✓

$$P_C = \sum m(1, 2, 4, 7, 8, 11, 13, 14)$$

$$= a \oplus b \oplus c \oplus p$$



MAGNITUDE COMPARATOR
It compares 2 no. and gives 3 output
 $a > b$, $a = b$ and $a < b$.

1 Bit Magnitude Comparator

i/p		output			
a	b	$a > b$	$a = b$	$a < b$	
0	0	0	1	0	$f_{a > b} = a \cdot \bar{b}$
0	1	0	0	1	$f_{a = b} = a \odot b$
1	0	1	0	0	$f_{a < b} = \bar{a} \cdot b$
1	1	0	1	0	

2 Bit Magnitude Comparator

It compares 2 bits of a + b

	a ₁	a ₀	b ₁	b ₀	$f_{a > b}$	$f_{a = b}$	$f_{a < b}$
0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1
4	0	1	0	0	1	0	0
5	0	1	0	1	0	1	0
6	0	1	1	0	0	0	1
7	0	1	1	1	0	0	1
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	1	0
11	1	0	1	1	0	0	1
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	1	0

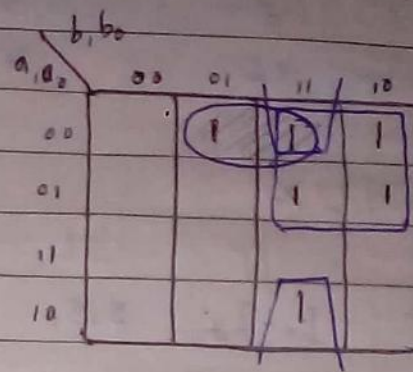
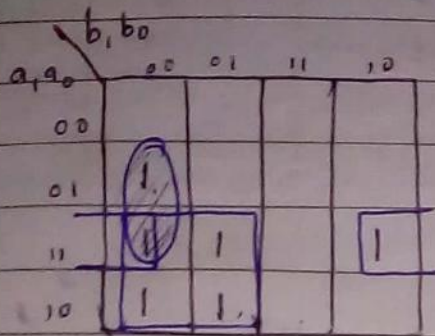
Date: / / 201

Page No.

$$f_{a>b} = \sum m(4, 8, 9, 12, 13, 14) = a_1 \bar{b}_1 + a_0 b_1 \bar{b}_0 + a_1 a_0 \bar{b}_0$$

$$f_{a=b} = \sum m(0, 5, 10, 15)$$

$$f_{a<b} = \sum m(1, 2, 3, 6, 7, 11) = \bar{a}_1 b_1 + \bar{a}_1 \bar{a}_0 b_0 + \bar{a}_0 b_1 b_0$$



$$f_{a=b} = \sum m = (0, 5, 10, 15)$$

$$= \bar{a}_1 \bar{a}_0 \bar{b}_1 \bar{b}_0 + \bar{a}_1 a_0 \bar{b}_1 b_0 + a_1 \bar{a}_0 b_1 \bar{b}_0 + a_1 a_0 b_1 b_0$$

$$= \bar{a}_1 \bar{b}_1 (\bar{a}_0 \bar{b}_0 + a_0 b_0) + a_1 b_1 (\bar{a}_0 \bar{b}_0 + a_0 b_0)$$

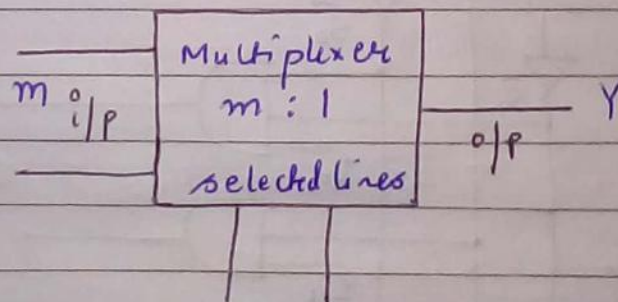
$$= \bar{a}_1 \bar{b}_1 (a_0 \odot b_0) + a_1 b_1 (a_0 \odot b_0)$$

$$= (\bar{a}_1 \bar{b}_1 + a_1 b_1) (a_0 \odot b_0)$$

$$= (a_1 \odot b_1) (a_0 \odot b_0)$$

MULTIPLEXER

$m = 2^s$
 $m = \text{no. of i/p}$
 $s = \text{selected lines}$



It has many i/p and only one o/p and it connects only one i/p with o/p depending upon the value of selected lines. That's why it is also called data selector or many to one logic circuit or 11el to serial converter.

MUX	i/p s	slines	o/p
2:1	2	1	L
4:1	4	2	L
8:1	8	3	L
16:1	16	4	L

4:1 MUX

	i/p	sl	o/p
4:1	$I_0 - I_3$	S_1, S_0	Y

$S_1 \quad S_0 \quad Y$

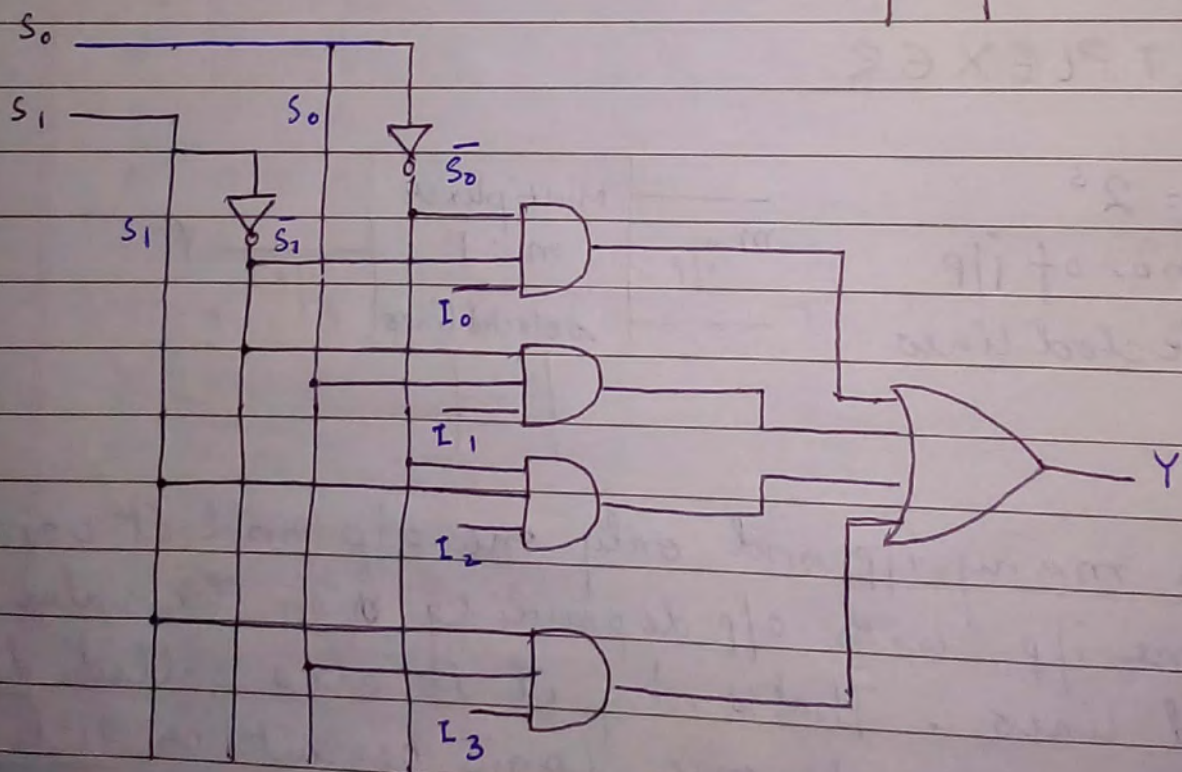
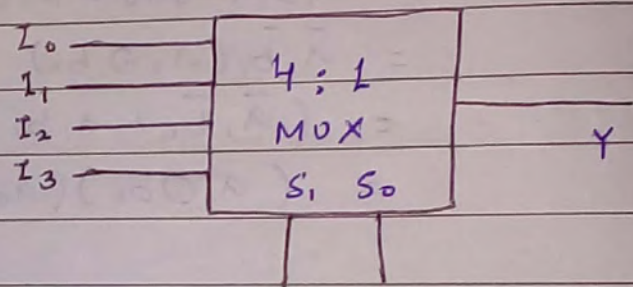
0 0 I_0

0 1 I_1

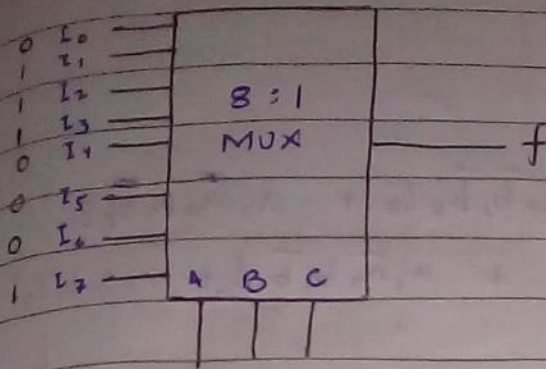
1 0 I_2

1 1 I_3

$$Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$



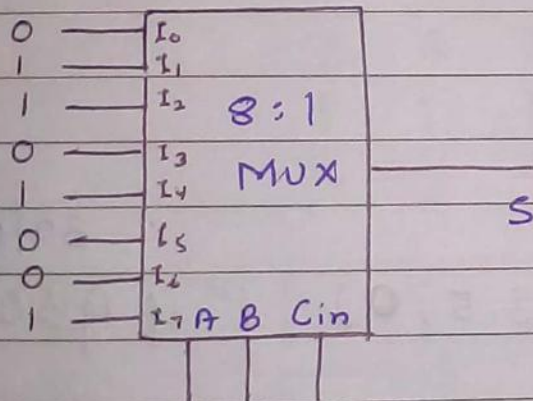
Q. Implement $f = \sum m(1, 2, 3, 7)$ by using 8:1 MUX



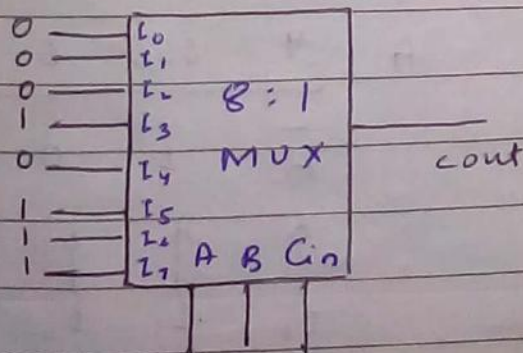
$$f = \bar{A}\bar{B}CI_1 + \bar{A}B\bar{C}I_2 + \bar{A}BCI_3 + ABCI_7$$

In Multiplexer, output is in S.O.P. form and all the minterms given in f are connected to +5V and remaining terms are connected to GND and only one output is possible in one time

Q. Implement full adder by using 8:1 MUX
 $f(A, B, C_{in}) = \sum m(1, 2, 4, 7)$

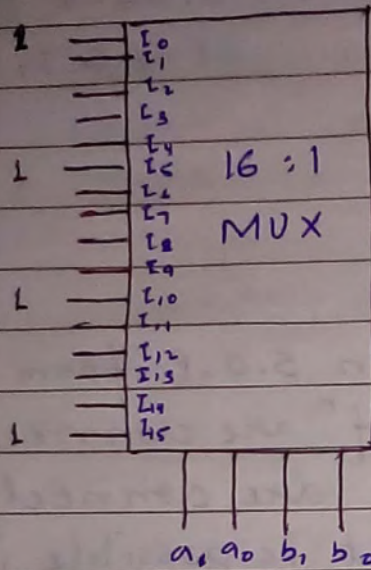


$$S = ABC_{in}I_1 + A\bar{B}C_{in}I_2 + \bar{A}BC_{in}I_4 + ABC_{in}I_7$$



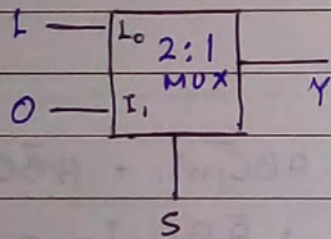
$$cout = A\bar{B}\bar{C}_{in}I_3 + \bar{A}B\bar{C}_{in}I_5 + \bar{A}B\bar{C}_{in}I_6 + ABC_{in}I_7$$

Q. Implement $f_{a=b}$ of 2 bit Magnitude comparator by using 16:1 MUX.



$$f_{a=b} = \bar{a}_1 \bar{a}_0 \bar{b}_1 \bar{b}_0 I_0 + \bar{a}_1 \bar{a}_0 b_1 b_0 I_5 + a_1 \bar{a}_0 b_1 \bar{b}_0 I_{10} + a_1 a_0 b_1 b_0 I_{15}$$

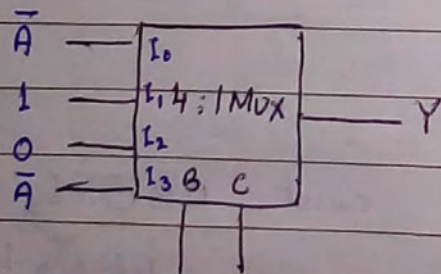
Q. Implement NOT gate using 2:1 MUX



$$Y = \bar{S} I_0 + S I_1$$

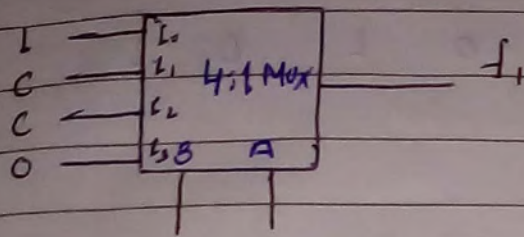
30/08/17

Q. Implement $f_1 = \sum m(1, 3, 5, 7)$, f^n of ABC



$\bar{A}$	(0)	(1)	2	(3)
A	4	(5)	6	7
$\bar{A}$	1	0	$\bar{A}$	
$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	
I_0	I_1	I_2	I_3	

$\bar{C}$	(0)	2	4	6
C	(1)	(3)	(5)	7
	$\uparrow L$	$\uparrow C$	$\uparrow C$	$\uparrow 0$
	I_0	I_1	I_2	I_3

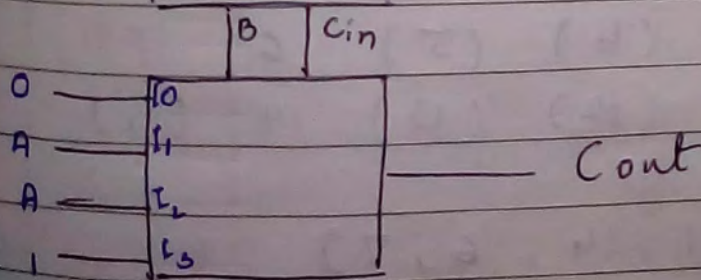
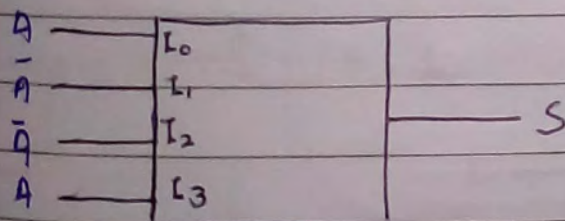


Q. Implement full adder by using 4:1 mux.
(if B + C are selected lines)

$$S = \sum m(1, 2, 4, 7)$$

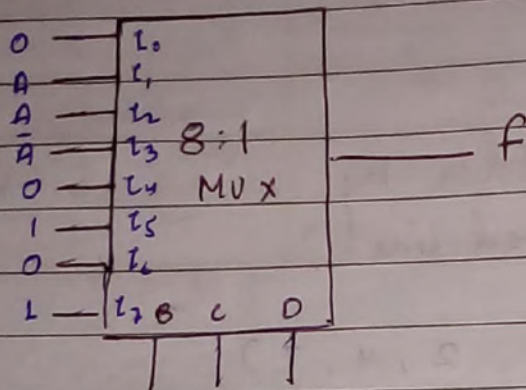
$$Cout = \sum m(3, 5, 6, 7)$$

$\bar{A}$	0	(1)	(2)	3
A	(4)	5	6	(7)
	A	$\bar{A}$	$\bar{A}$	A
	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
	I_0	I_1	I_2	I_3

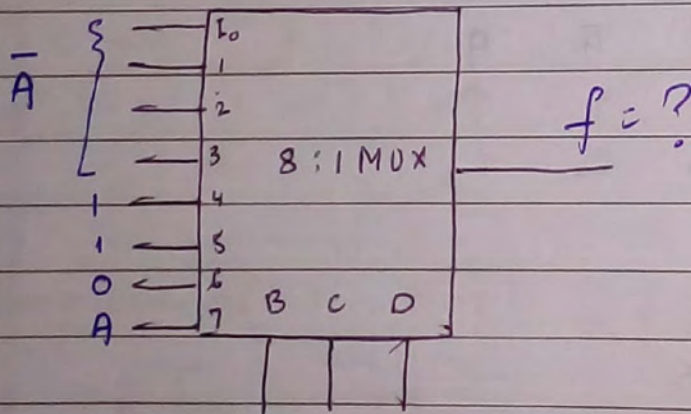


Q. Implement $f = \sum m(3, 5, 7, 9, 10, 13, 15)$ by using 8:1 MUX

$\bar{A}$	0	1	2	(3)	4	(5)	6	(7)
A	8	(9)	(10)	11	12	(13)	14	(15)
	0	A	A	$\bar{A}$	0	1	0	1



Q. In the following MUX circuit, what will be value of f in maxterm



$\bar{A}$	(0)	(1)	(2)	(3)	(4)	(5)	6	7
A	8	9	10	11	(12)	(13)	14	(15)

$$f = \prod M(8, 9, 10, 11, 14, 6, 7)$$

Q. Minimum no. of 2: MUX req. to implement Half Adder.

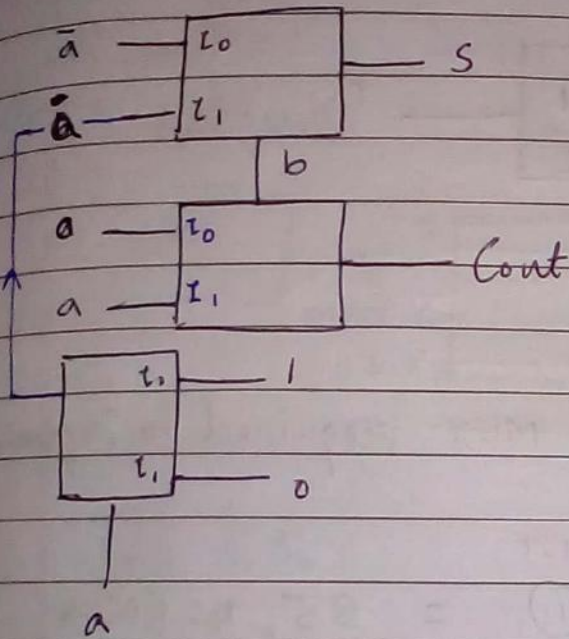
A	B
0	0
0	1
1	0
1	1

$$S = \sum m(1, 2)$$

$$Cout = \sum m(3)$$

0	(1)
(2)	3
a	$\bar{a}$

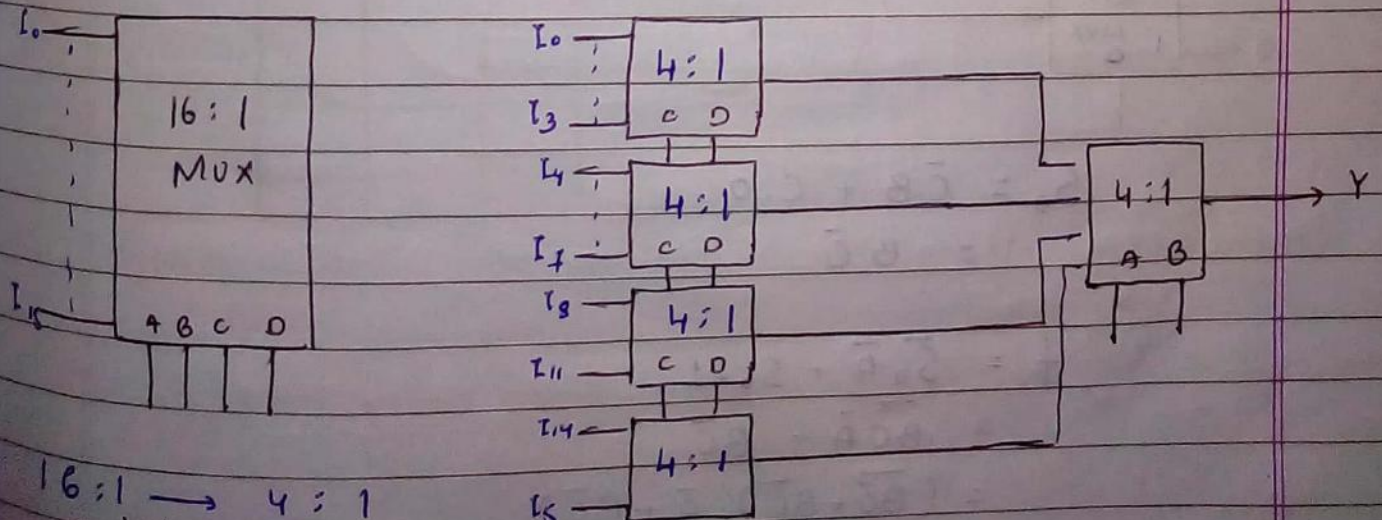
0	1
2	(3)
0	1



3, 2: MUX required

MULTIPLEXER TREE

Q. Implement 16:1 MUX by using 4:1 MUX



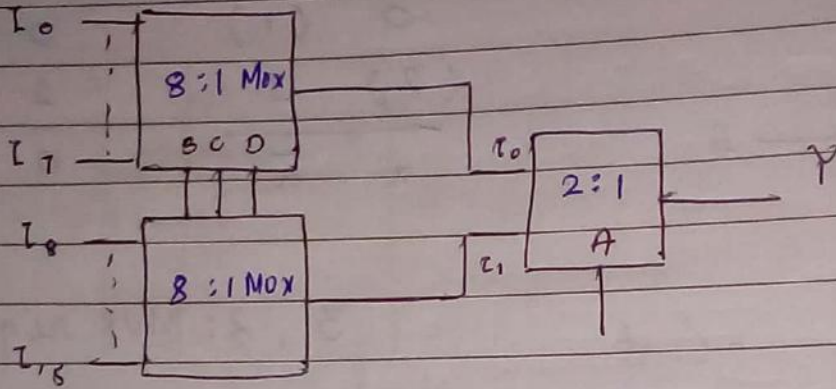
$$16:1 \rightarrow 4:1$$

$$\frac{16}{4} = 4 \quad \frac{4}{4} = 1 \quad = 5$$

Q. Implement 16:1 MUX by using 8:1 MUX and 2:1 MUX

$$16:1 \rightarrow 8:1$$

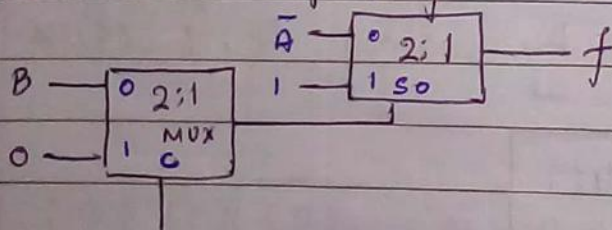
$$\frac{16}{8} = \frac{2}{2} = \frac{1}{1}$$



Q. Determine total no. of 4:1 MUX required to implement 256:1 MUX

$$\frac{256}{4} = \frac{64}{4} + \frac{16}{4} + \frac{4}{4} + \frac{1}{4} = 85, 4:1 \text{ MUX}$$

Q. Determine output f in the following circuit



$$S_0 = \bar{C}B + C \cdot 0$$

$$= B\bar{C}$$

$$f = S_0 \bar{A} + S_0 \cdot 1$$

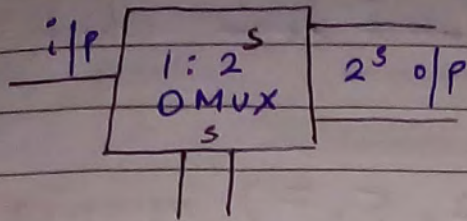
$$= B\bar{C}\bar{A} + B\bar{C}$$

$$= (B\bar{C} + B\bar{C})(\bar{A} + 1)$$

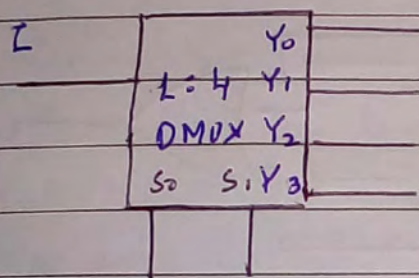
$$= \bar{A} + B\bar{C}$$

DEMULTIPLEXER (DEMUX):

Reverse process of MUX. It has only one i/p and many o/p



1:4 DEMUX



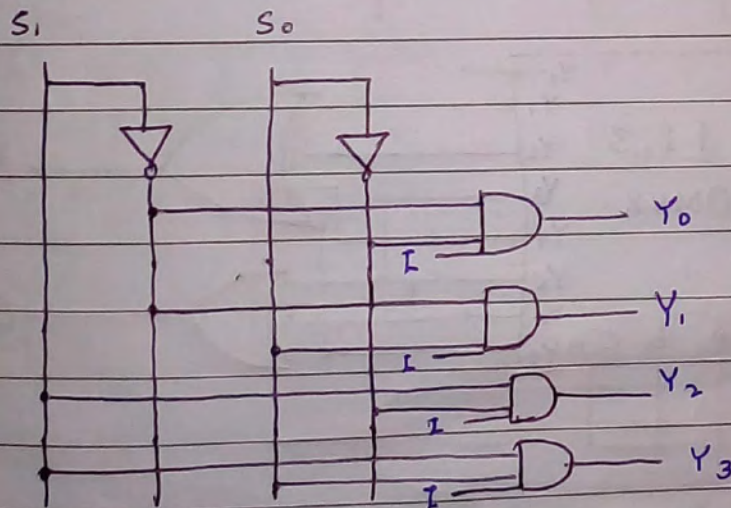
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

$$Y_0 = \bar{S}_1 \bar{S}_0 I$$

$$Y_1 = \bar{S}_1 S_0 I$$

$$Y_2 = S_1 \bar{S}_0 I$$

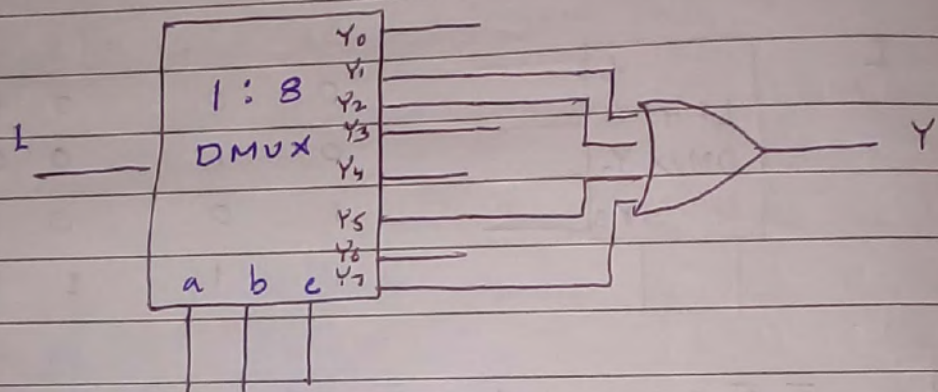
$$Y_3 = S_1 S_0 I$$



IMPLEMENTATION of any f^n using DMUX

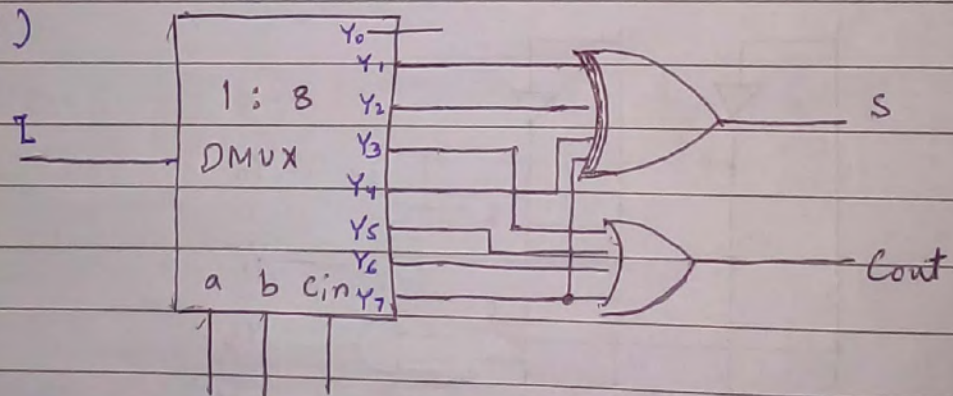
Connect all the minterms (outputs) to input of OR gate which are given in f^n and leave remaining minterms and leave which are not given in f^n .

Q. Implement $Y = \sum m(1, 2, 5, 7)$ by using 1:8 DMUX and this f^n is dependent upon ABC variable.



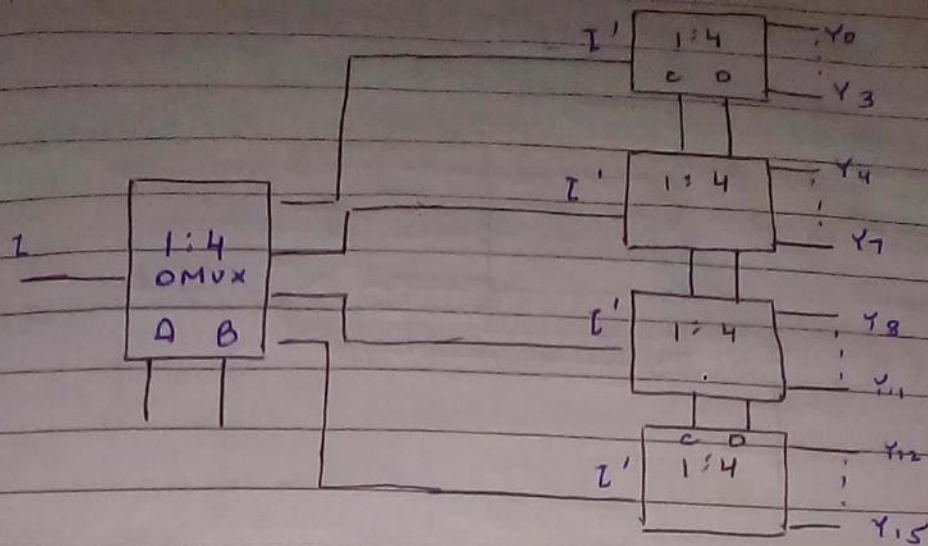
Q. Implement full adder using 1:8 DMUX

$$S = \sum m(1, 2, 4, 7)$$



Q. Implement 1:16 DMUX by using 1:4 DMUX

$$\frac{16}{4} = \frac{4}{4} = 1 = 5 \text{ 1:4 DMUX required.}$$



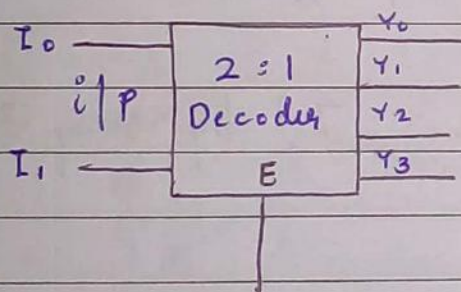
DECODER

It converts m lines to 2^m lines, i.e.

It has m inputs and 2^m outputs.

It has one enable line. Decoder will only work if enable line is activated. e.g. 2:4, 3:8, 4:16
5:32

2:4 DECODER



E	I ₁	I ₀	Y ₃	Y ₂	Y ₁	Y ₀
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

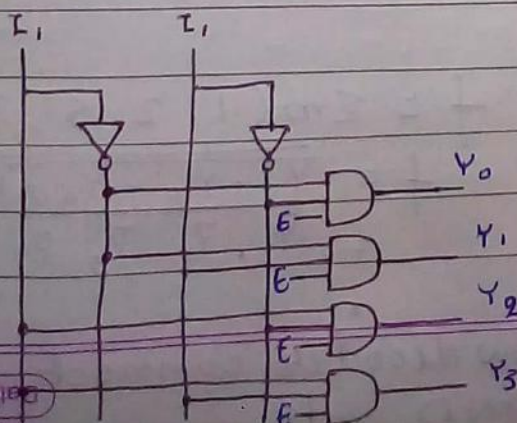
Active High 2:4 Decoder
0 x x Not Working

$$Y_0 = \bar{I}_0 \bar{I}_1 E$$

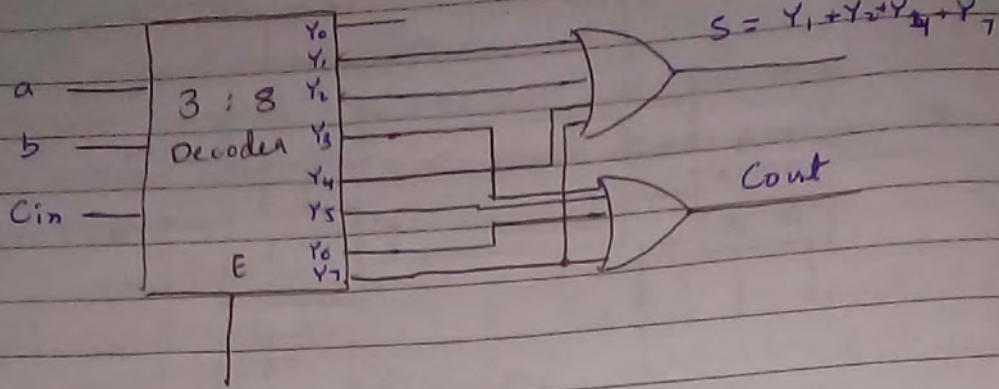
$$Y_1 = \bar{I}_1 I_0 E$$

$$Y_2 = I_1 \bar{I}_0 E$$

$$Y_3 = I_1 I_0 E$$

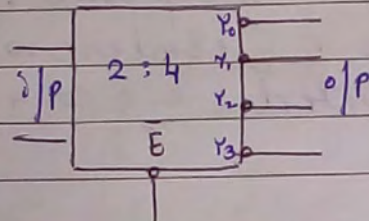


Q. Implement full adder by using 3:8 Decoder.



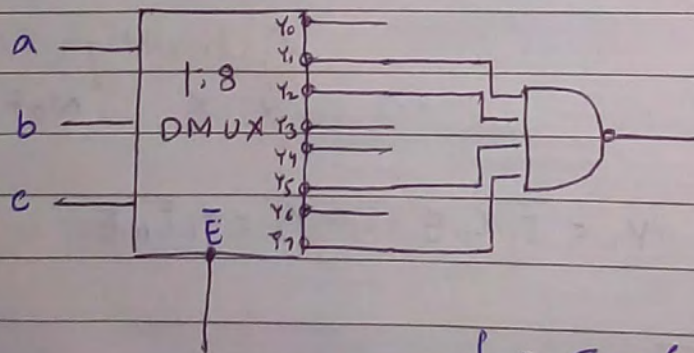
2:4 Active LOW DECODER

In this decoder will only work when value of enable line is 0.



$\bar{E}$	I_1	I_0	Y_3	Y_2	Y_1	Y_0
0	0	0	1	1	1	0
0	0	1	1	1	0	1
0	1	0	1	0	1	1
0	1	1	0	1	1	1
1	x	x	not working.			

Q. Implement $f = \sum m(1, 2, 5, 7)$ by using 3:8 active low decoder.



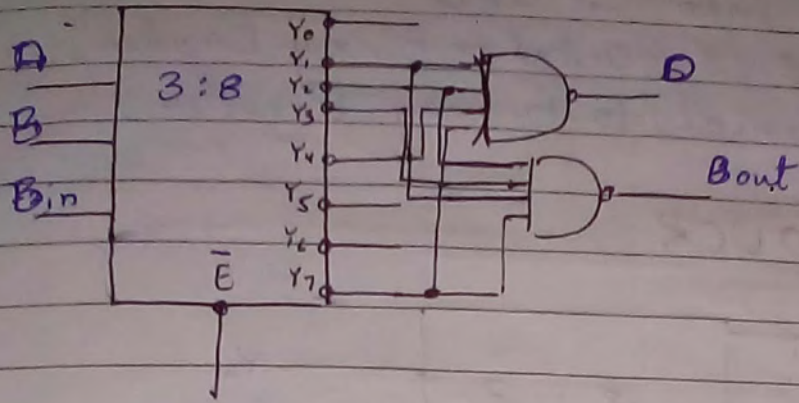
$$f = \sum m(1, 2, 5, 7)$$

$$f = \overline{Y_1 + Y_2 + Y_5 + Y_7}$$

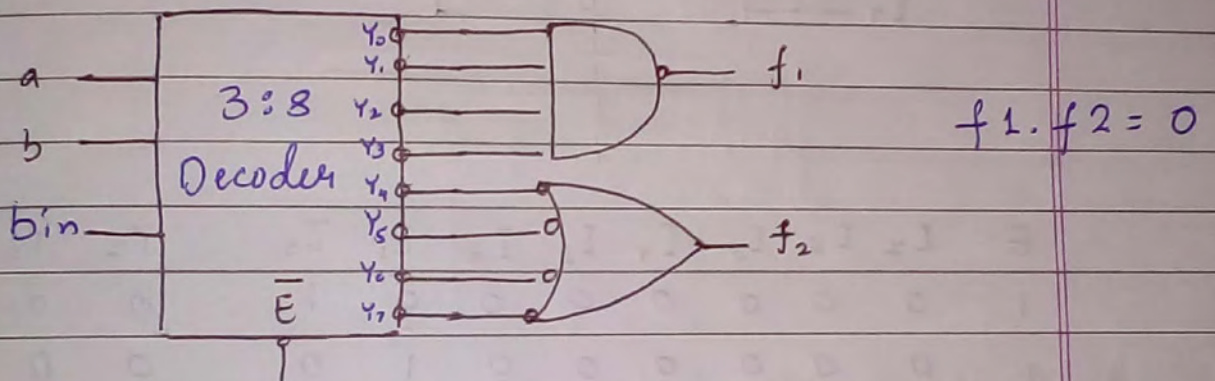
$$= \overline{Y_1} \cdot \overline{Y_2} \cdot \overline{Y_5} \cdot \overline{Y_7}$$

In active low decoder connect output with NAND gate

Q. Implement full subtracter by using active low 3:8 decoder. 01/09/17

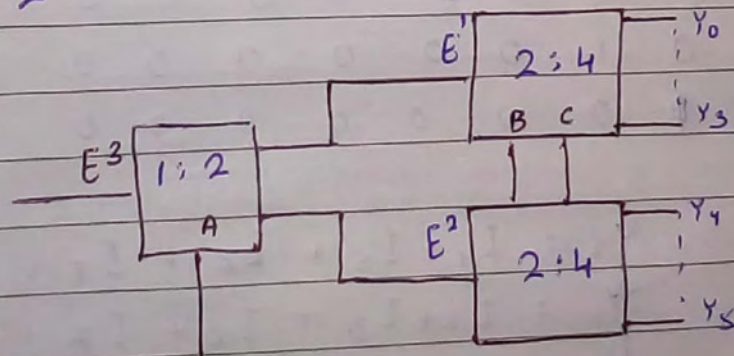


Q. What will be value of f_1 and f_2 in the following circuit



Q. Implement 3:8 decoder by using 2:4 and 1:2 decoder

$$\frac{8}{4} = \frac{(2)}{2} = (1)$$

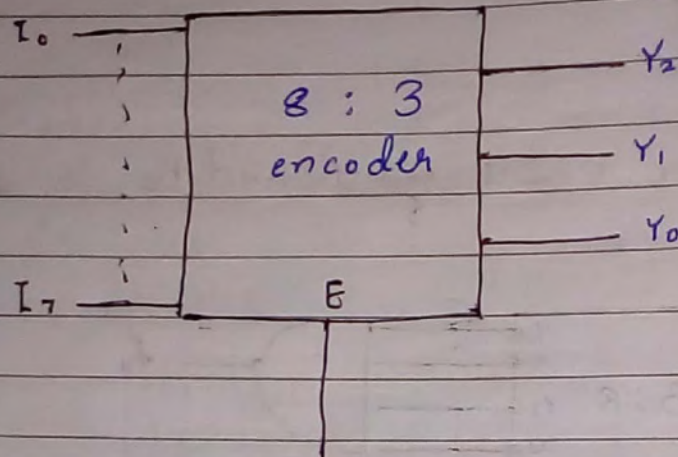


ENCODER

Reverse process of decoder. It converts 2^m lines into m lines.

e.g. 4:2, 8:3 (Octal to Binary Encoder), 16:4
(Hexadecimal to Binary Encoder).

8:3 ENCODER

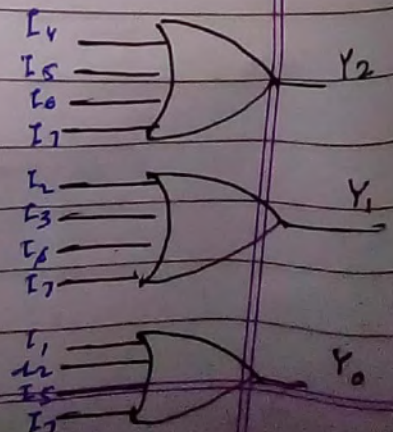


E	I ₇	I ₆	I ₅	I ₄	I ₃	I ₂	I ₁	I ₀	Y ₂	Y ₁	Y ₀
1	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	1	0	0	0	1
1	0	0	0	0	0	1	0	0	0	1	0
1	0	0	0	0	1	0	0	0	0	1	1
1	0	0	0	1	0	0	0	0	1	0	0
1	0	0	1	0	0	0	0	0	1	0	1
1	0	1	0	0	0	0	0	0	1	1	0
1	1	0	0	0	0	0	0	0	1	1	1

$$Y_2 = I_4 + I_5 + I_6 + I_7$$

$$Y_1 = I_2 + I_3 + I_6 + I_7$$

$$Y_0 = I_1 + I_3 + I_5 + I_7$$



PRIORITY ENCODER

In octal to binary encoder, if at one time, more than one i/p are high or if all i/p are low we do not get any output and this problem is avoided by priority encoder. In this we set highest priority to that no. which has highest binary value.

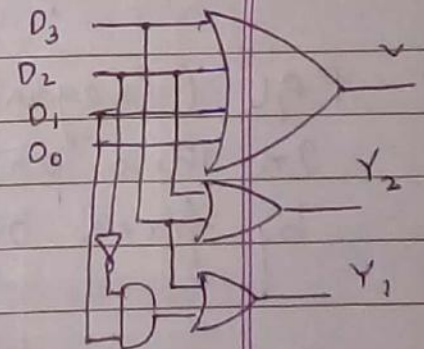
D_3	D_2	D_1	D_0	Y_2	Y_1	Y
1	x	x	x	1	1	1
0	1	x	x	1	0	1
0	0	1	x	0	1	1
0	0	0	1	0	0	1
0	0	0	0	x	x	0

$$Y = \pi M(0) = D_3 + D_2 + D_1 + D_0$$

$$Y_2 = \pi M(1, 2, 3) + \sum d(0)$$

$$Y_1 = \pi M(1, 4, 5, 6, 7) + \sum d(0)$$

$D_3 D_2$	$D_1 D_0$			
	00	01	11	10
00	d_0	d_1	2	2
01	d_4	d_5	d_7	d_6
11	12	13	15	14
10	8	9	11	10



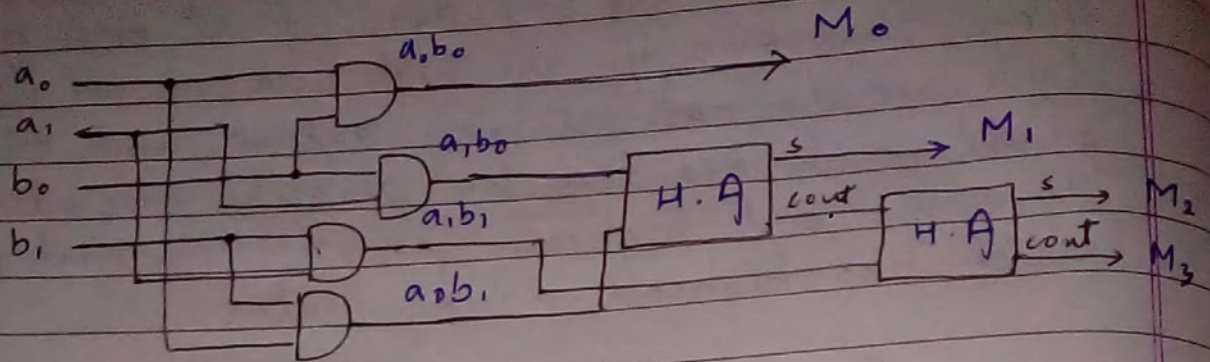
$$Y_1 = (D_3 + D_1)(D_3 + \bar{D}_2)$$

$$= D_3 + D_1 \bar{D}_2$$

$D_3 D_2$	$D_1 D_0$			
	00	01	11	10
00	d_0	d_1	d_2	d_3
01				
11				
10				

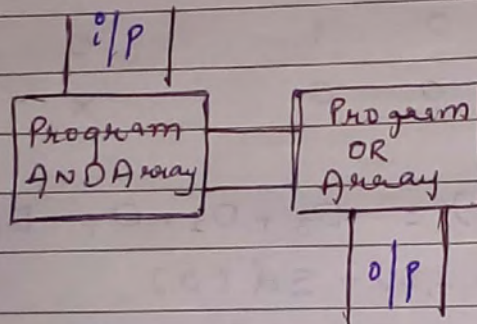
$$Y_2 = D_3 + D_2$$

2 BIT BINARY MULTIPLIER



PLA (Programmable LOGIC Array)

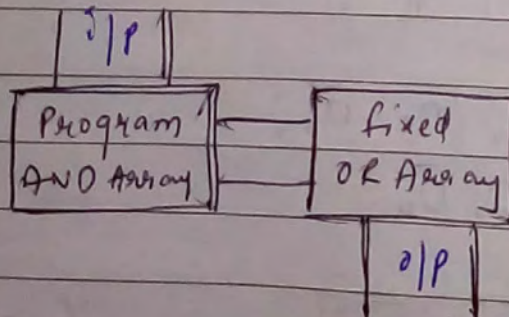
In this both AND & OR array are programmable



AND arrays are followed by OR arrays.

PAL (Programmable array LOGIC)

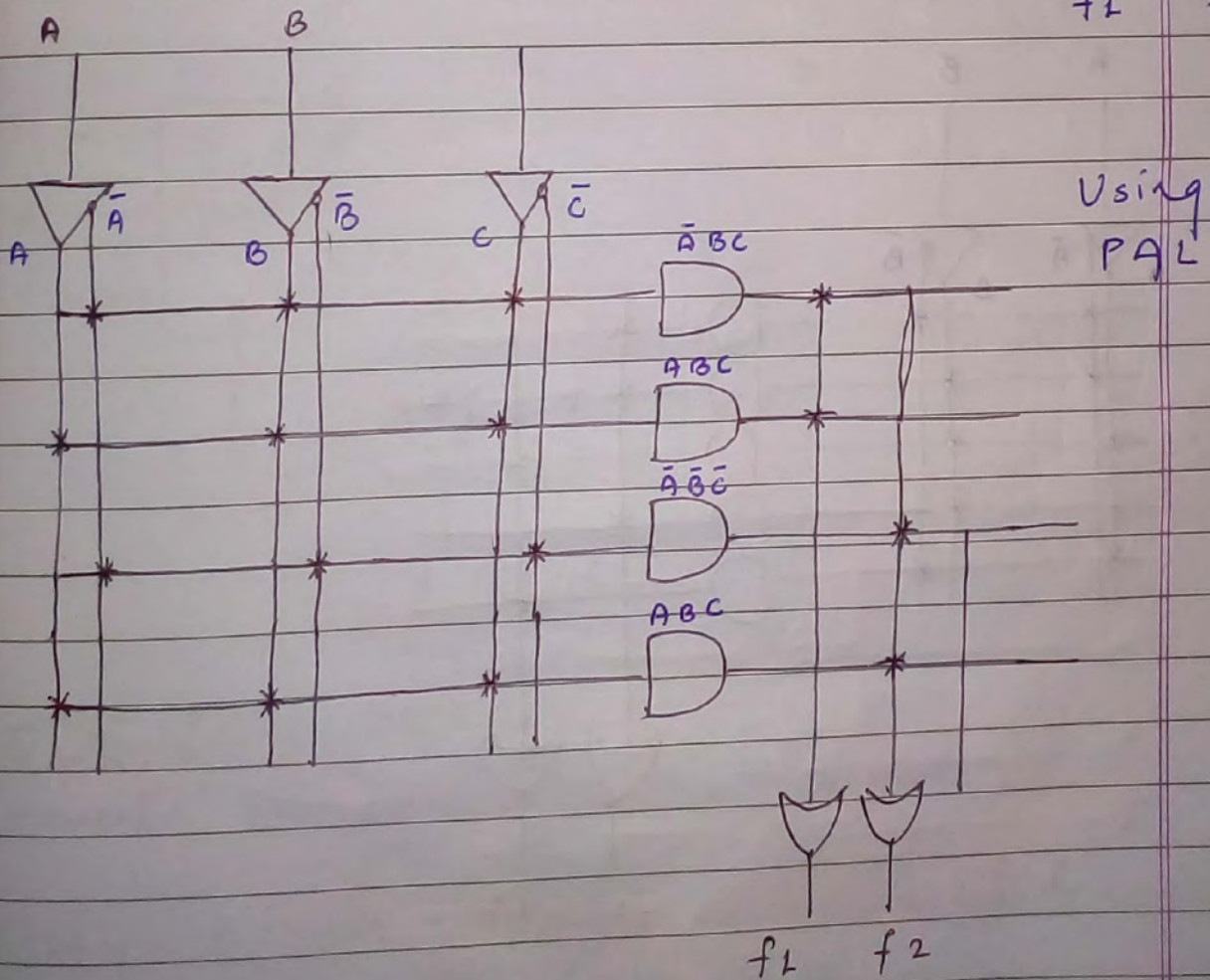
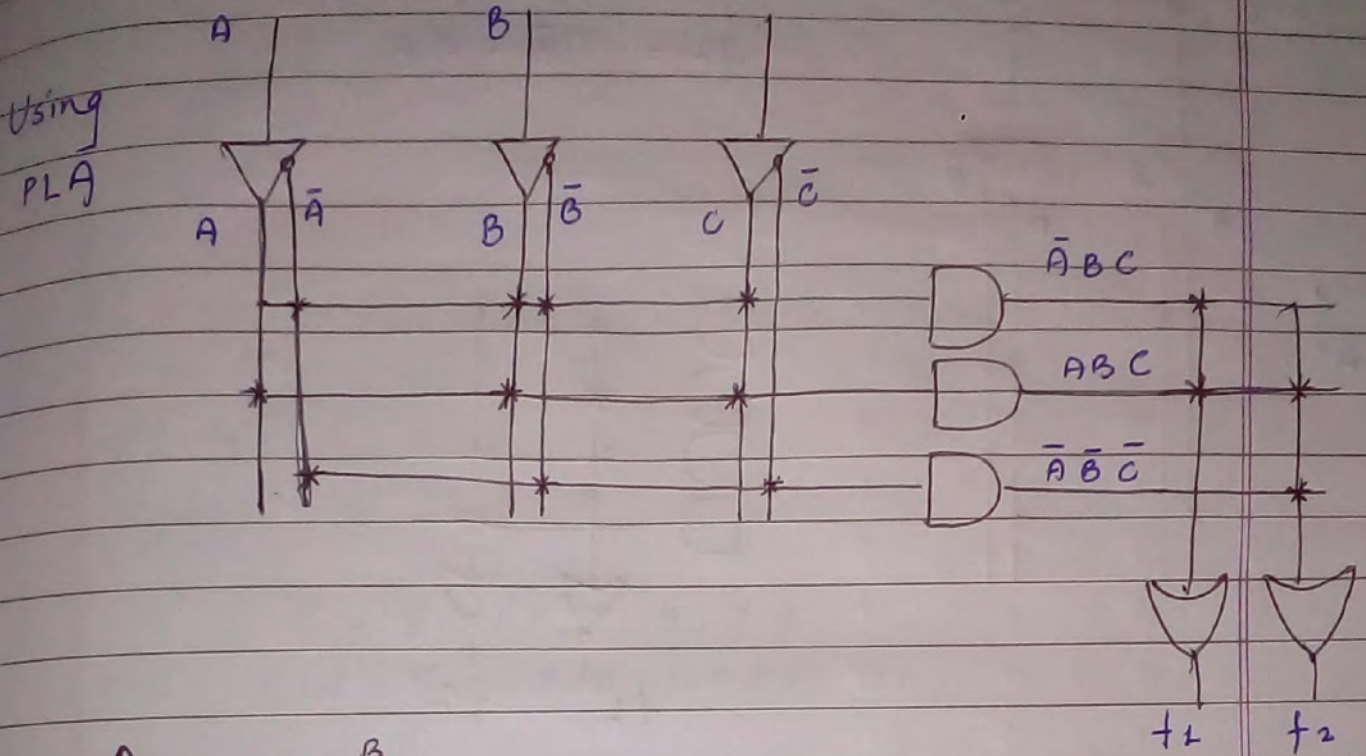
In this programmable AND arrays are followed by fixed OR arrays



Q. Implement the following f' by using PLA & PAL

$$f_1 = \bar{A}BC + ABC$$

$$f_2 = \bar{A}\bar{B}\bar{C} + ABC$$

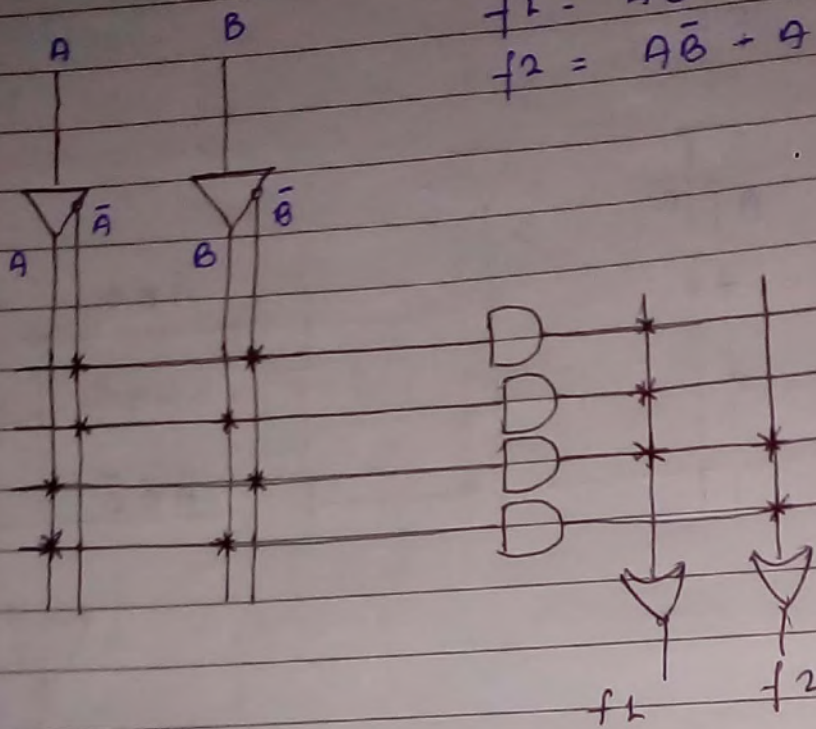


Q. Implement $f_1 = \sum m(0, 1, 2)$ by PLA
 $f_2 = \sum m(2, 3)$

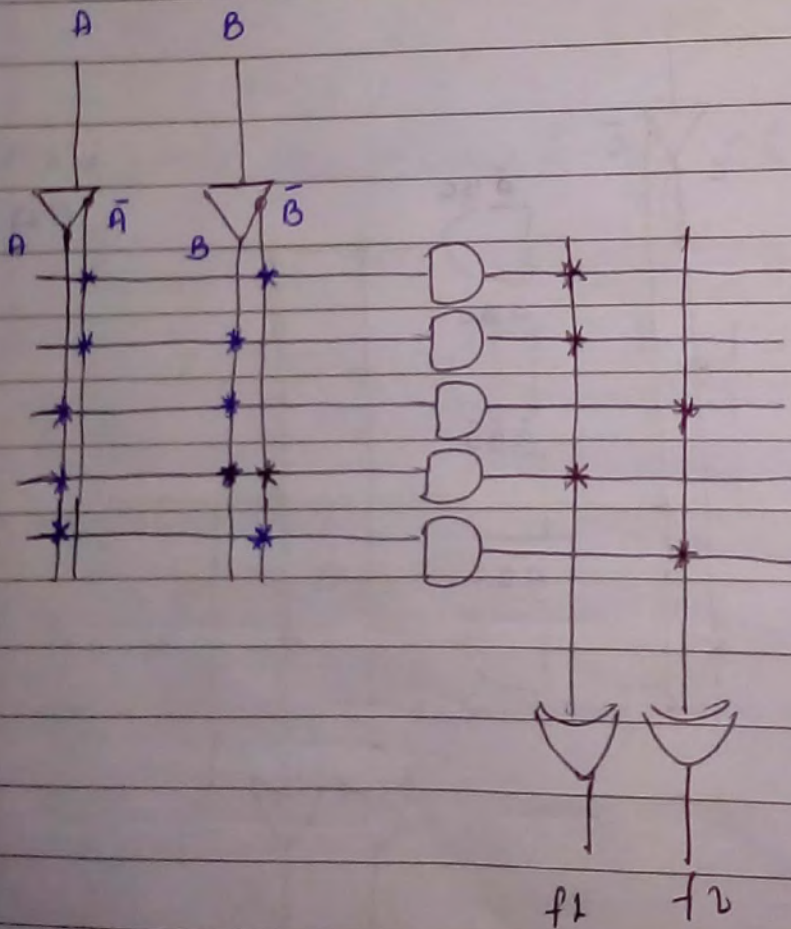
$$f_1 = \bar{A}\bar{B} + \bar{A}B + A\bar{B}$$

$$f_2 = A\bar{B} + AB$$

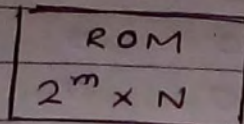
Using
PLA



Using
PAL



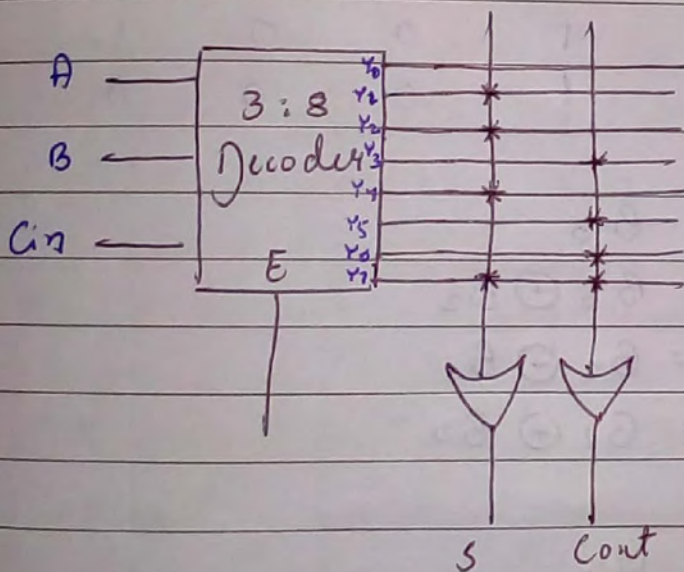
E-PROM [ROM]



m = total address lines
 N = length of data bus

Q. Design full adder by using ROM
In ROM both AND arrays and OR arrays are fixed and with the help of decoder we can implement any f^n .

$$S = \sum m(1, 2, 4, 7)$$
$$Cont = \sum m(0, 3, 5, 6, 7)$$



CODE CONVERTER

Converts one code into another code.

Q. Design a code converter of 4 Bits which converts binary code into Grey code

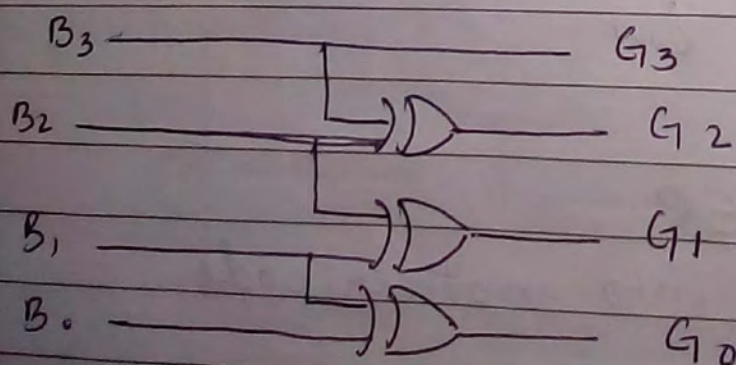
B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

$$G_3 = B_3$$

$$G_2 = B_3 \oplus B_2$$

$$G_1 = B_2 \oplus B_1$$

$$G_0 = B_1 \oplus B_0$$



SEVEN SEGMENT CODE :

BCD	a	b	c	d	e	f	g
0	1	1	1	1	1	1	0
1	0	1	1	0	0	0	0
2	1	1	0	1	1	0	1
3	1	1	1	1	0	0	1
4	0	1	1	0	0	1	1
5	1	0	1	1	0	1	1
6	1	0	1	1	1	1	1
7	1	1	1	0	0	0	0
8	1	1	1	1	1	1	1
9	1	1	1	0	0	1	1

Q. Seven Segment has two push buttons P₁ and P₂, when a button is pressed the prize is displayed in Seven Segment display.

If no buttons are pressed 0 is displayed.

If P₁ is pressed but P₂ not 2 is displayed.

If P₂ is pressed only 5 is displayed.

If both are pressed there is error and it signifies by EE.

* If segment a to g are considered with option is correct

$$a) g = \bar{P}_1 + P_2, d = c + e$$

$$c) g = P_1 + P_2, d = c + e$$

$$b) g = P_1 + P_2, e = b + c$$

* Minimum no. of NOT & OR gates are required to design this display.

P ₁	P ₂	a	b	c	d	e	f	g
0	0	1	1	1	1	1	1	0
0	1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	0	1
1	1	1	0	0	1	1	1	1

$$\begin{aligned}
 a &= \underline{1} \\
 b &= \underline{p_2} \\
 c &= \underline{p_1} \\
 e &= \underline{p_1 + p_2} \\
 f &= \underline{p_1} + p_2 \\
 g &= p_1 + p_2
 \end{aligned}$$

$$\begin{aligned}
 d &= \overline{p_2 + p_1 + p_2} \\
 d &= 1 + p_1 = 1
 \end{aligned}$$

3 NOT 3 OR